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ESCUELA DE CIENCIAS Y SISTEMAS
SISTEMAS ORGANIZACIONALES Y GERENCIALES 1
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Manual de Instalación

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GUATEMALA 26 DE MARZO DEL AÑO 2026

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1. Manuales de Instalación

a. Manual de Instalación del ERP — Odoo Community Edition (vía Docker)

a. Requerimientos del Sistema

Requisito	Especificación Mínima
Sistema Operativo	Ubuntu 22.04 LTS
RAM	4 GB mínimo (8 GB recomendado)
CPU	2 núcleos mínimo
Disco	40 GB de espacio libre
Docker Engine	Docker 24.x o superior
Docker Compose	Plugin docker compose (v2) integrado en Docker Engine
Navegador	Chrome, Firefox o Edge (última versión)
Acceso a Internet	Requerido para descargar imágenes de Docker Hub

Ventaja del enfoque Docker:

No es necesario instalar Python, PostgreSQL ni dependencias del sistema manualmente. Docker empaqueta todo el entorno de Odoo (servidor de aplicación, base de datos y dependencias) en contenedores aislados y reproducibles, garantizando que el entorno sea idéntico en cualquier máquina.

Paso 1: Actualizar el sistema e instalar dependencias previas

[illegible]

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
-o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] \
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
sudo apt update
```

[illegible]

```
sudo apt install -y docker-ce docker-ce-cli containerd.io \
docker-buildx-plugin docker-compose-plugin
```

[illegible]

```
docker --version  
docker compose version
```

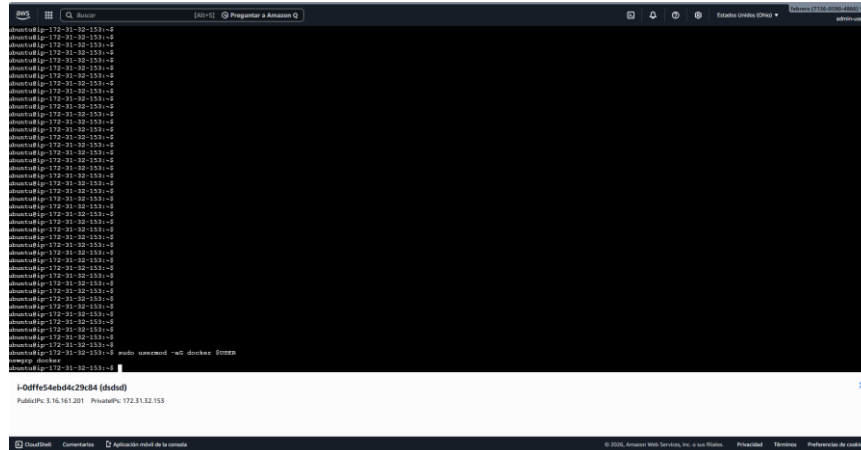
Salida esperada:

```
Docker version 26.1.4, build 5650f9b
Docker Compose version v2.27.1
```

[illegible]

Paso 5: Agregar el usuario actual al grupo docker

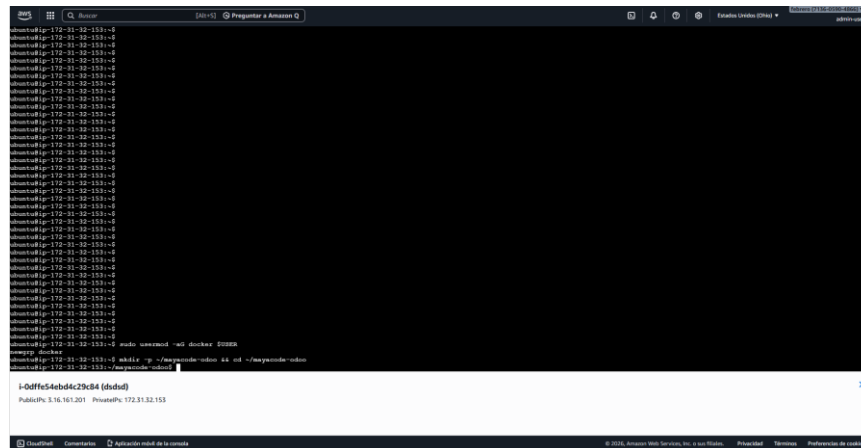
```
sudo usermod -aG docker $USER  
newgrp docker
```



A screenshot of a terminal window with a dark background. The prompt is root@ip-172-31-32-153:~#. The terminal shows a list of directory listings for /root/. At the bottom, the command `sudo usermod -aG docker $USER` has been executed, followed by `newgrp docker`. The terminal output shows the command was successful. Below the terminal window, a status bar indicates the public IP is 3.16.161.201 and the private IP is 172.31.32.153.

Paso 6: Crear el directorio de trabajo del proyecto

```
mkdir -p ~/mayacode-odoo/addons  
cd ~/mayacode-odoo
```



A screenshot of a terminal window with a dark background. The prompt is root@ip-172-31-32-153:~#. The terminal shows the command `mkdir -p ~/mayacode-odoo/addons` being executed. Below this, the command `cd ~/mayacode-odoo` is shown. The terminal output shows the directory was created successfully. Below the terminal window, a status bar indicates the public IP is 3.16.161.201 and the private IP is 172.31.32.153.

Paso 7: Crear el archivo docker-compose.yml

```
cat << 'EOF' > docker-compose.yml
version: '3.8'

services:

  db:
    image: postgres:15
    container_name: mayacode_postgres
    restart: always
    environment:
      POSTGRES_DB: postgres
      POSTGRES_USER: odoo
      POSTGRES_PASSWORD: MayaCode2026
    volumes:
      - mayacode_pgdata:/var/lib/postgresql/data
    networks:
      - mayacode_net

  odoo:
    image: odoo:17.0
    container_name: mayacode_odoo
    restart: always
    depends_on:
      - db
    ports:
      - "8069:8069"
    environment:
      HOST: db
      USER: odoo
      PASSWORD: MayaCode2026
    volumes:
      - mayacode_odoo_data:/var/lib/odoo
      - ./addons:/mnt/extra-addons
    networks:
      - mayacode_net

volumes:
  mayacode_pgdata:
  mayacode_odoo_data:

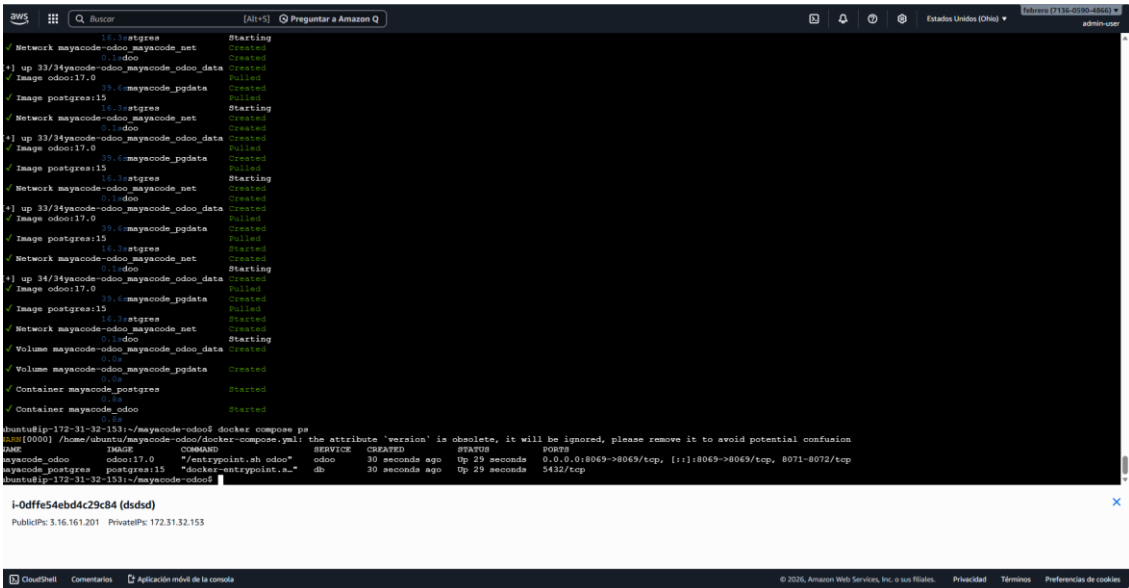
networks:
  mayacode_net:
    driver: bridge
EOF
```


Paso 10: Verificar contenedores activos

```
docker compose ps
```

Salida esperada:

NAME	IMAGE	STATUS	PORTS
mayacode_odoo	odoo:17.0	Up	0.0.0.0:8069->8069/tcp
mayacode_postgres	postgres:15	Up	5432/tcp

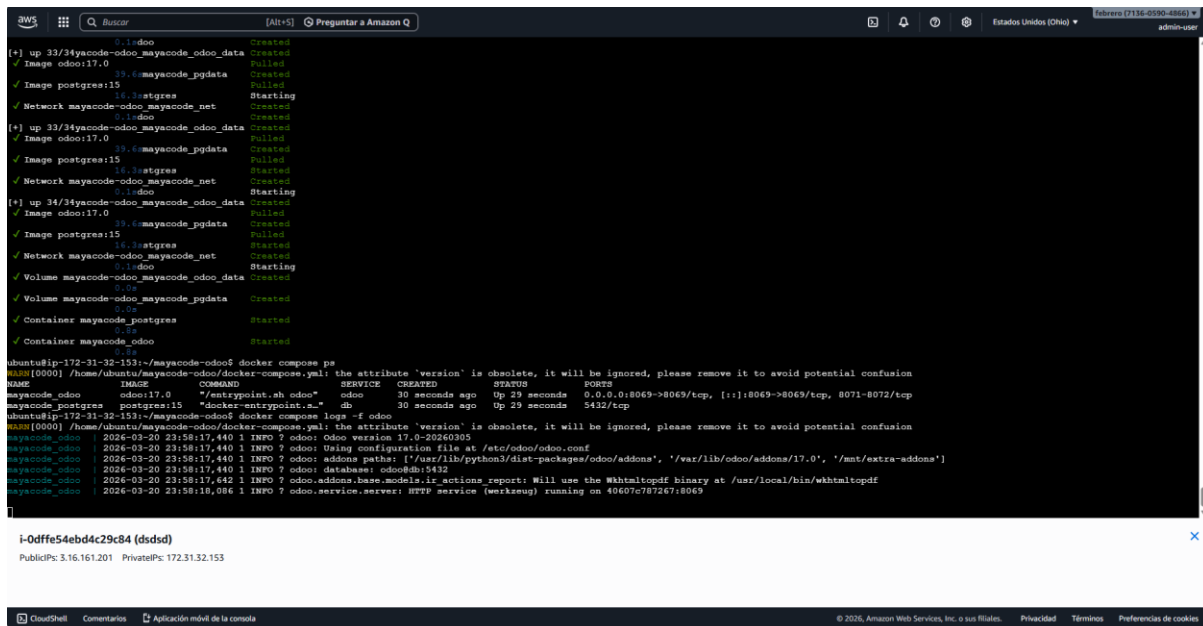


Paso 11: Verificar logs de Odoo

```
docker compose logs -f odoo
```

Mensaje esperado:

INFO odoo.service.server: HTTP service (werkzeug) running on 0.0.0.0:8069



```
0.1.odoo Created
[+] up 33/34yacode-odoo_mayacode_odoo_data Created
Image odoo:17.0 Pulled
33.6.mayacode_pgdata Created
Image postgres:15 Pulled
Network mayacode-odoo_mayacode_net Starting
0.1.odoo Created
[+] up 33/34yacode-odoo_mayacode_odoo_data Created
Image odoo:17.0 Pulled
33.6.mayacode_pgdata Created
Image postgres:15 Pulled
Network mayacode-odoo_mayacode_net Starting
0.1.odoo Created
[+] up 34/34yacode-odoo_mayacode_odoo_data Created
Image odoo:17.0 Pulled
33.6.mayacode_pgdata Created
Image postgres:15 Pulled
Network mayacode-odoo_mayacode_net Starting
0.1.odoo Created
Volume mayacode-odoo_mayacode_odoo_data Created
Volume mayacode-odoo_mayacode_pgdata Created
Container mayacode_postgres Started
Container mayacode_odoo Started

ubuntu@ip-172-31-32-153:~/mayacode-odoo$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
mayacode_odoo odoo:17.0 "/entrypoint.sh odoo" odoo 30 seconds ago Up 29 seconds 0.0.0.0:8069->8069/tcp, [::]:8069->8069/tcp, 8071-8072/tcp
mayacode_postgres postgres:15 "docker-entrypoint.s..." db 30 seconds ago Up 29 seconds 5432/tcp

ubuntu@ip-172-31-32-153:~/mayacode-odoo$ docker compose logs -f odoo
INFO[0000] /home/ubuntu/mayacode-odoo/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
mayacode_odoo 2026-03-20 23:58:17.440 I INFO ? odoo: Odoo version 17.0-20260305
mayacode_odoo 2026-03-20 23:58:17.440 I INFO ? odoo: Using configuration file at /etc/odoo/odoo.conf
mayacode_odoo 2026-03-20 23:58:17.440 I INFO ? odoo: addons paths: ['/usr/lib/python3/dist-packages/odoo/addons', '/var/lib/odoo/addons/17.0', '/mnt/extra-addons']
mayacode_odoo 2026-03-20 23:58:17.642 I INFO ? odoo: database: odoo@db:5432
mayacode_odoo 2026-03-20 23:58:18.086 I INFO ? odoo.addons.base.models.ir_actions_report: Will use the Wkhtmltopdf binary at /usr/local/bin/wkhtmltopdf
mayacode_odoo 2026-03-20 23:58:18.086 I INFO ? odoo.service.server: HTTP service (werkzeug) running on 0.0.0.0:8069
```

i-0dffe54ebd4c29c84 (dsdsd)

PublicIPs: 3.16.161.201 PrivateIPs: 172.31.32.153

CloudShell Comentarios Aplicación móvil de la consola

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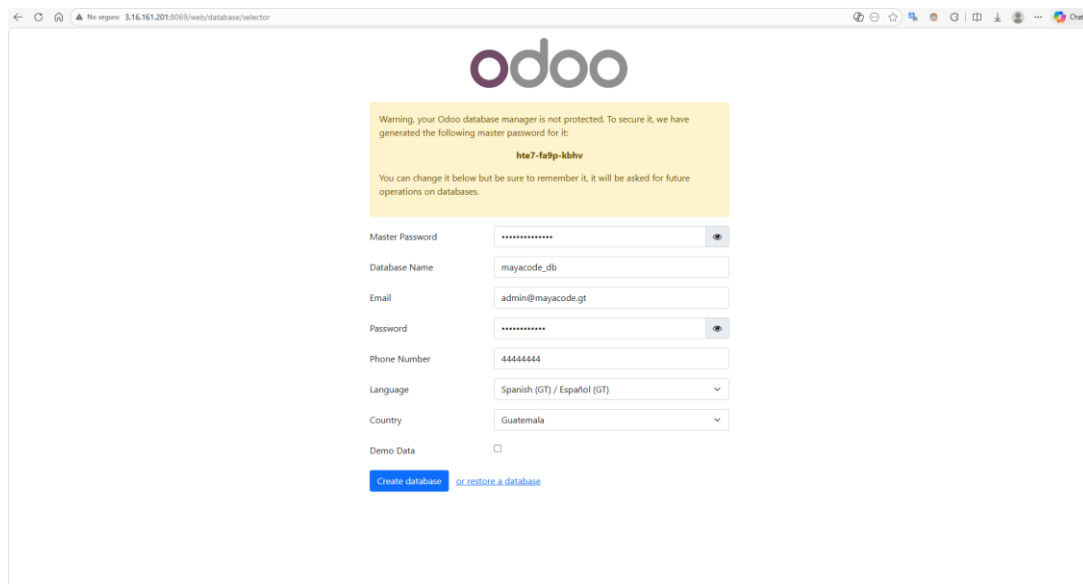
c. Validación Final del ERP

Acceder a:

http://{direccion IP}:8069

Datos de creación de base de datos:

Campo	Valor
Database Name	mayacode_db
Email	admin@mayacode.gt
Password	MayaCode2026
Language	Español
Country	Guatemala



Warning, your Odoo database manager is not protected. To secure it, we have generated the following master password for it:

htz7-fa3p-kbhv

You can change it below but be sure to remember it, it will be asked for future operations on databases.

Master Password:

Database Name:

Email:

Password:

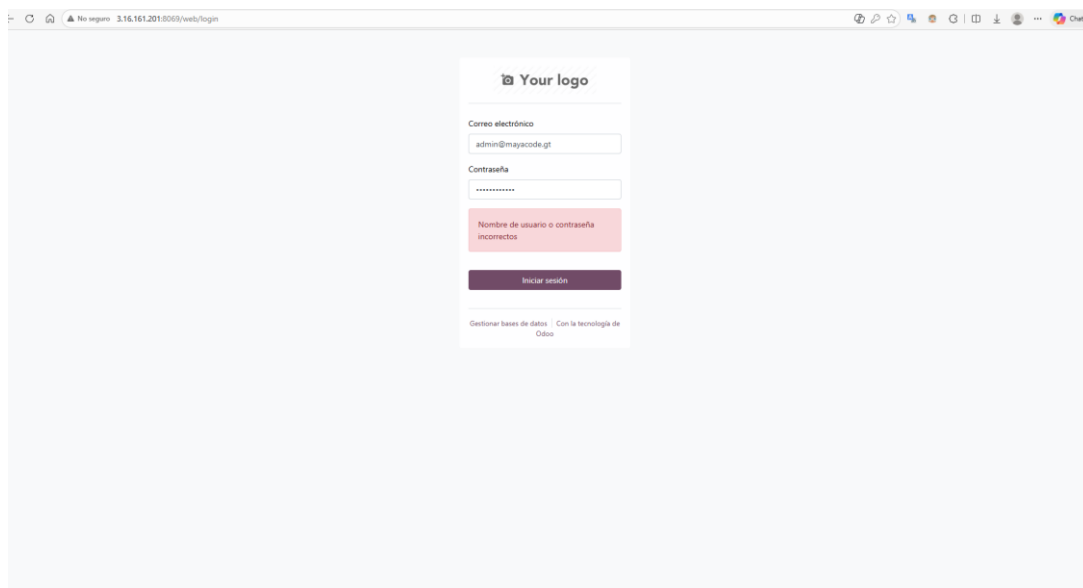
Phone Number:

Language:

Country:

Demo Data: ☐

[Create database](#) [or restore a database](#)





Correo electrónico:

Contraseña:

Nombre de usuario o contraseña incorrectos

[Iniciar sesión](#)

Gestionar bases de datos Con la tecnología de Odoo

b. Manual de Activación del CRM — Odoo CRM (Módulo Integrado)

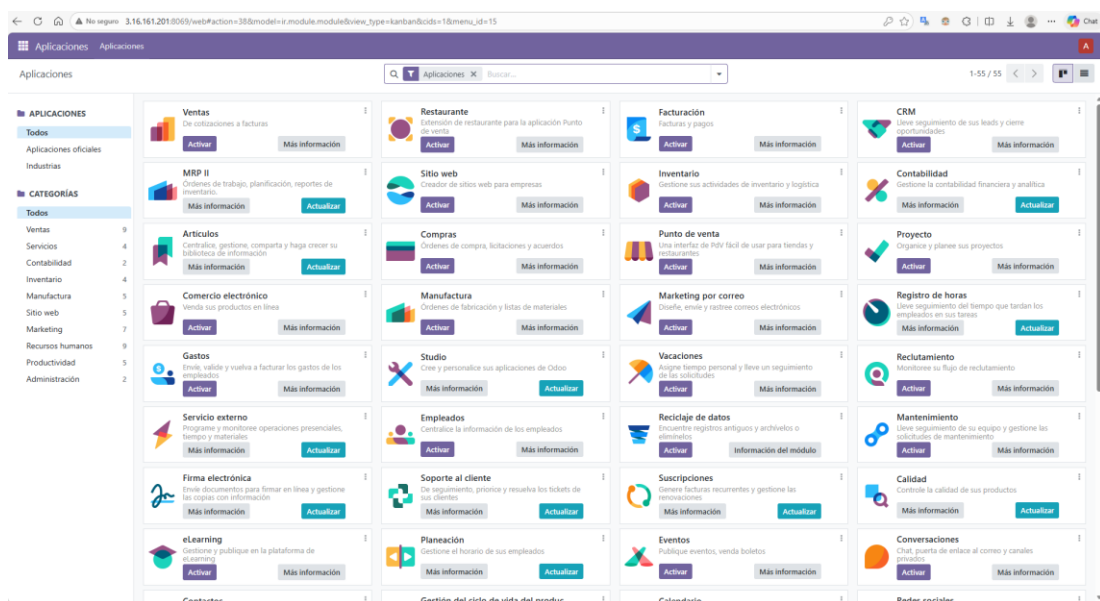
a. Requerimientos

Odoo CRM es un módulo nativo. No requiere instalación adicional.

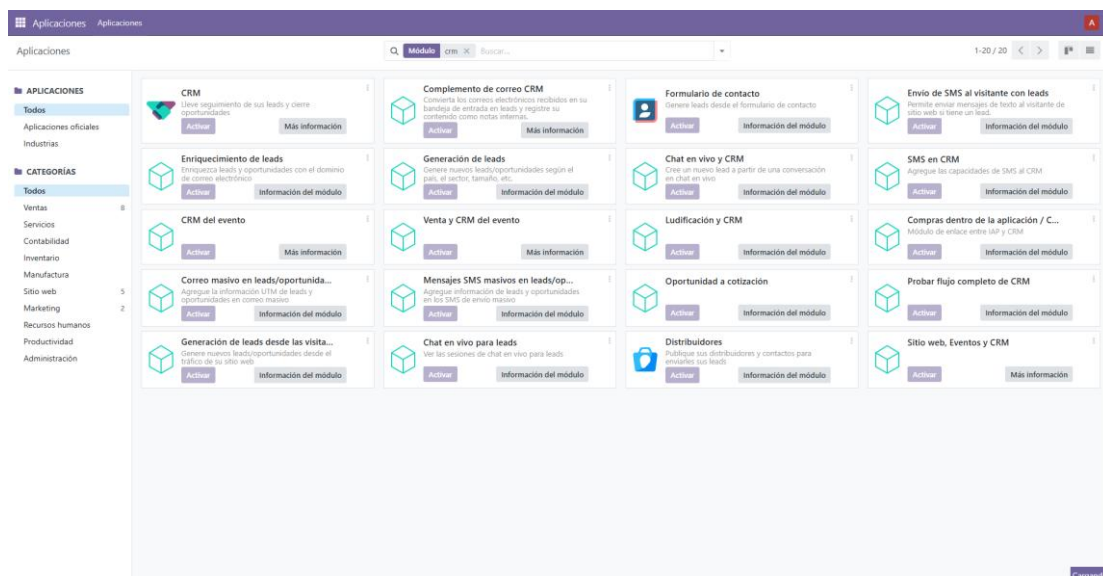
Prerequisito: Contenedores activos (docker compose ps).

b. Procedimiento de Activación

1. Ingresar a <http://localhost:8069> como administrador.
2. Ir a Apps.

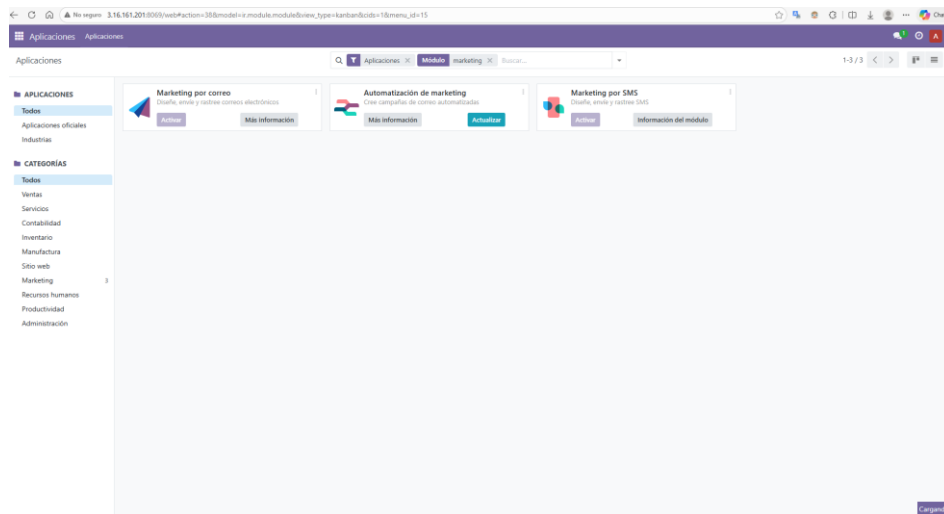


3. Buscar CRM (quitar filtro “Apps”).
4. Instalar CRM (Odoo S.A.).

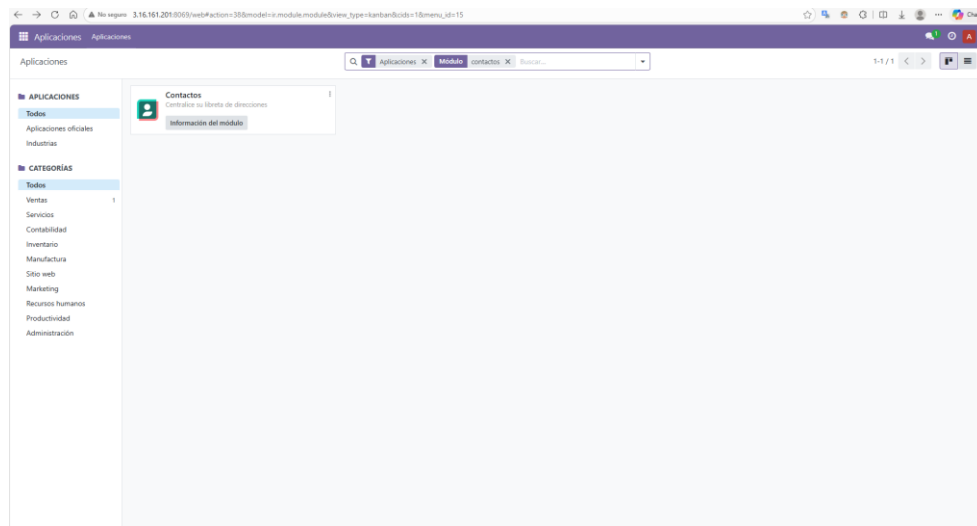


5. Instalar módulos complementarios:

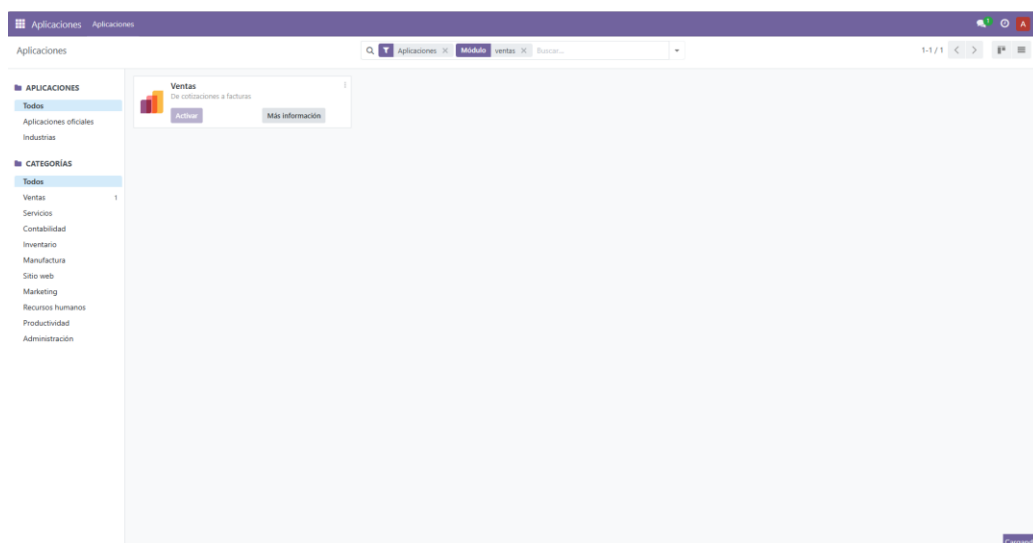
○ Email Marketing



○ Contactos

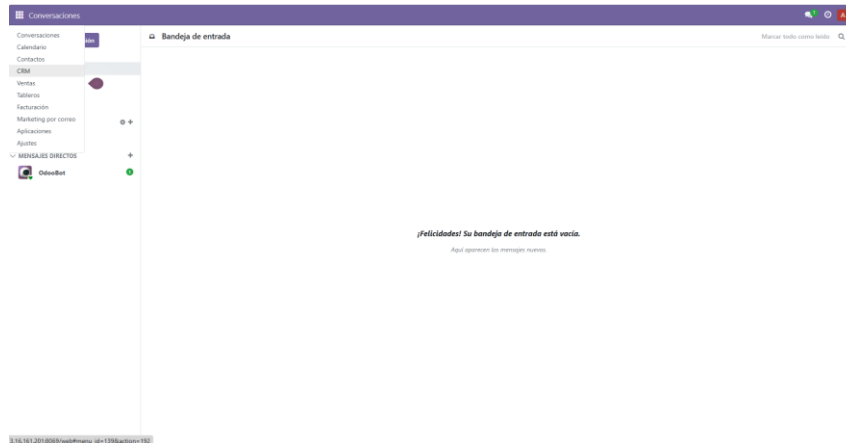


○ Ventas

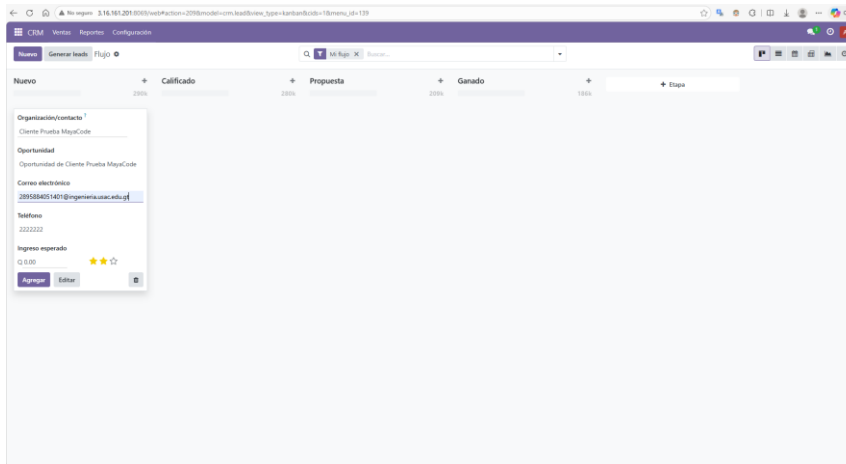


c. Validación Final del CRM

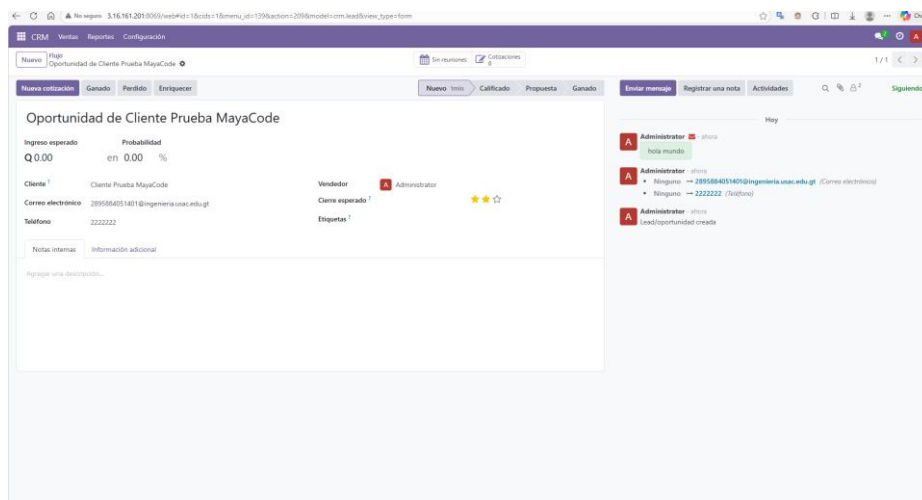
- Menú CRM visible.



- Pipeline Kanban: *New* → *Qualified* → *Proposition* → *Won*.
- Crear contacto: Cliente Prueba MayaCode.



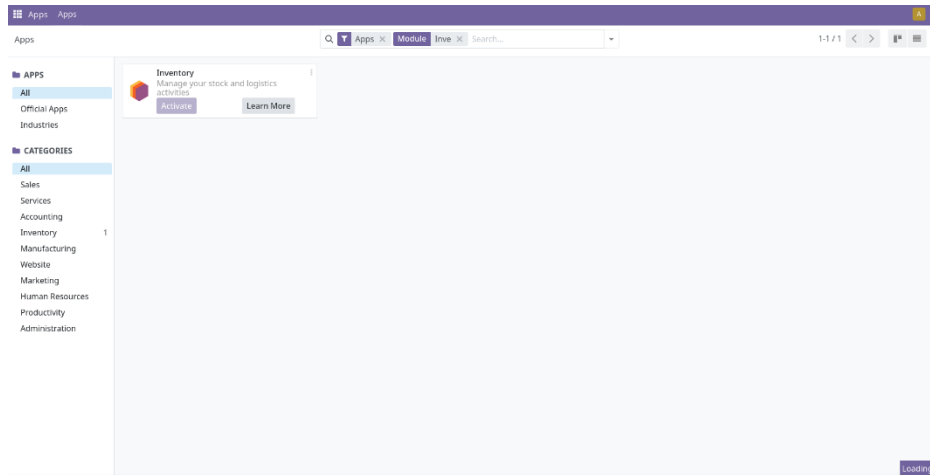
- Verificar Email Marketing operativo.



c. Bienvenido a Odoo

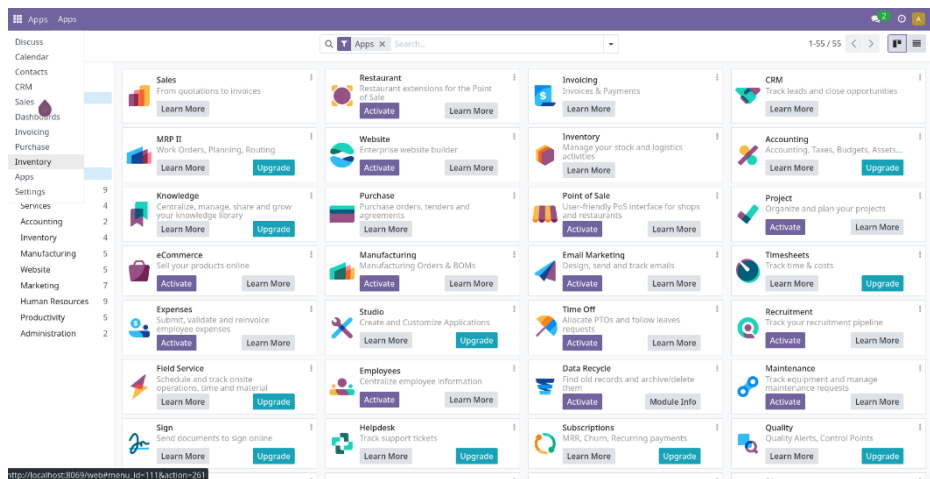
Dentro de la vista de Odoo, debemos instalar los siguientes módulos:

- Inventory
- Purchase
- Sales
- CRM
- Invoicing

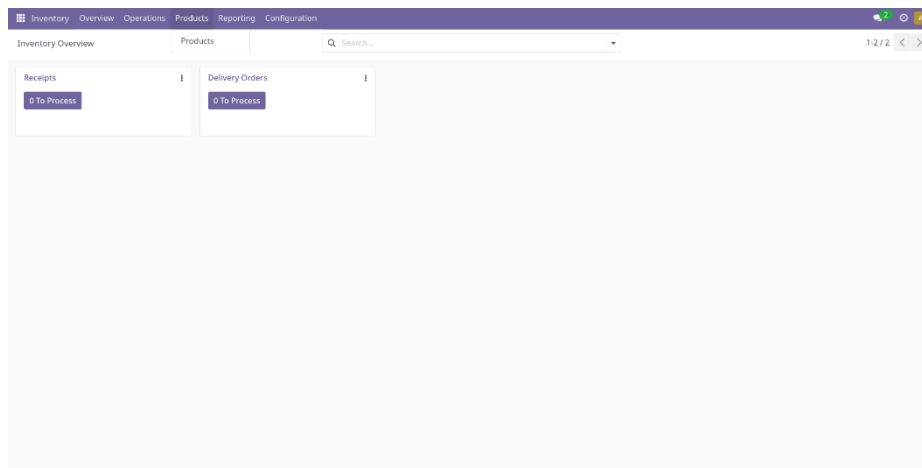


Carga masiva de Productos

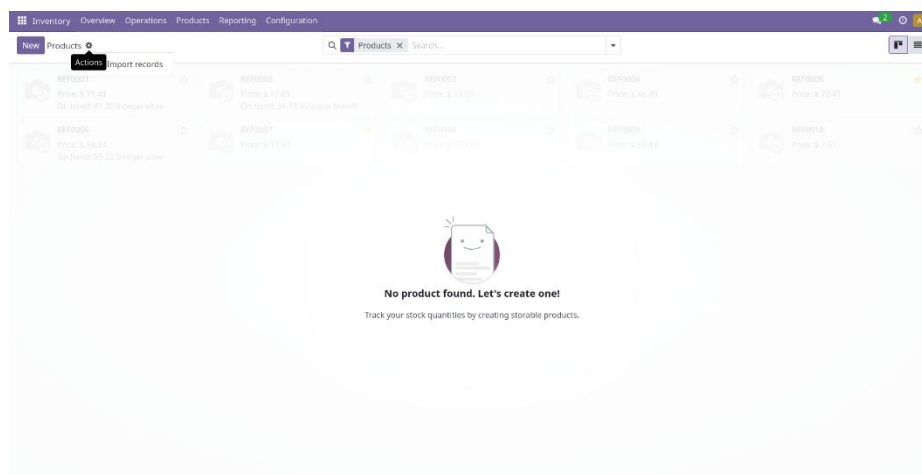
Seleccionamos el módulo de Inventory.



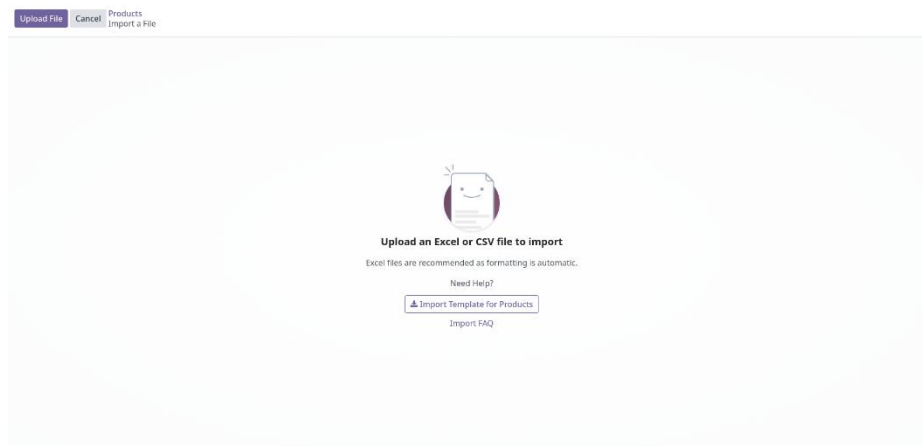
Luego en la parte superior seleccionamos Products.



En la tuerca que está a la derecha del buscador, seleccionamos Import records.



Y presionamos Upload File.



Luego solo mapeamos los campos correctamente y presionamos Import.

Inventory Overview Operations Products Reporting Configuration

Import Test Load File Cancel Products Import a File

products.csv

☒ Use first row as header

Formatting

Encoding: utf-8

Separator: Comma

Text Delimiter: "

Date Format: ?

Datetime Format: ?

Thousands Separator: Comma

Decimals Separator: Dot

Reimport

Help

Import Template for Products

Go to Import FAQ

name	Placas PCB personalizables	Ab	Name	x
default_code	EI-001	Ab	Internal Reference	x
type	product		Type	x
list_price	15.00	1.5	Sales Price	x
standard_price	8.00	1.5	Cost	x
categ_id/name	Microcontroladores	2=	Product Category	x
uom_id/name	Unidades	2=	Unit of Measure	x
uom_po_id/name	Unidades	2=	Unit of Measure	x
description_sale	Placa de circuito impreso personalizable. Ideal pa...	BP	Sales Description	x
sale_ok	TRUE	☒	Can be Sold	x
purchase_ok	TRUE	☒	Can be Purchased	x

Verificamos que los productos se carguen correctamente.

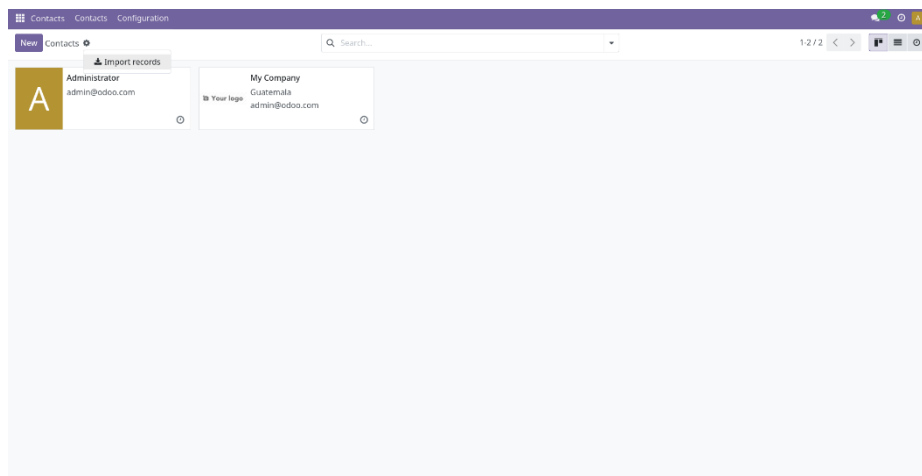
Arduino Uno R3 [EL-002] Price: 25.00 Q On hand: 0.00 Units	Bateria LiPo 3.7V 2000mAh [EL-012] Price: 8.00 Q On hand: 0.00 Units	Caja de proyecto ABS [EL-030] Price: 5.00 Q On hand: 0.00 Units	Camara OV2640 para ESP32-CAM [EL-015] Price: 6.00 Q On hand: 0.00 Units	Cargador TP4056 con protection [EL-013] Price: 2.00 Q On hand: 0.00 Units
Convertidor DC-DC Buck LM2596 [EL-011] Price: 3.50 Q On hand: 0.00 Units	Disipador aluminio 40x40x11mm [EL-020] Price: 2.50 Q On hand: 0.00 Units	Driver A4988 para stepper [EL-008] Price: 2.50 Q On hand: 0.00 Units	Estano sin plomo 0.8mm [EL-024] Price: 6.00 Q On hand: 0.00 Units	Fuente de alimentacion ATX 600W [EL-020] Price: 55.00 Q On hand: 0.00 Units
Matrix LED 8x8 MAX7219 [EL-016] Price: 5.50 Q On hand: 0.00 Units	Modulo CAN Bus MCP2515 [EL-027] Price: 7.00 Q On hand: 0.00 Units	Modulo ESP32 Wi-Fi BT [EL-004] Price: 8.00 Q On hand: 0.00 Units	Modulo GPS NEO-6M [EL-014] Price: 12.00 Q On hand: 0.00 Units	Modulo lector RFID RC522 [EL-019] Price: 4.00 Q On hand: 0.00 Units
Modulo rele 5V 4 canales [EL-009] Price: 5.00 Q On hand: 0.00 Units	Motor paso a paso NEMA 17 [EL-007] Price: 12.00 Q On hand: 0.00 Units	Multimetro digital DT-9205A [EL-022] Price: 18.00 Q On hand: 0.00 Units	Osciloscopio digital DSO138 [EL-021] Price: 35.00 Q On hand: 0.00 Units	Pantalla OLED 0.96 I2C [EL-005] Price: 4.50 Q On hand: 0.00 Units
Pasta termoconductor MX-4 [EL-025] Price: 8.50 Q On hand: 0.00 Units	Placas PCB personalizables [EL-001] Price: 15.00 Q On hand: 0.00 Units	Raspberry Pi 4 Model B [EL-003] Price: 45.00 Q On hand: 0.00 Units	Sensor corriente ACS712 30A [EL-028] Price: 6.50 Q On hand: 0.00 Units	Sensor de gas MQ-2 [EL-017] Price: 3.50 Q On hand: 0.00 Units
Sensor temperatura DHT22 [EL-010] Price: 4.00 Q On hand: 0.00 Units	Sensor ultrasonico HC-SR04 [EL-006] Price: 3.00 Q On hand: 0.00 Units	Servomotor SG90 9g [EL-018] Price: 3.00 Q On hand: 0.00 Units	Soldador de punta fina 60W [EL-023] Price: 22.00 Q On hand: 0.00 Units	Tira LED RGB WS2812B 1m [EL-029] Price: 9.00 Q On hand: 0.00 Units

Carga masiva de Clientes y Proveedores

Seleccionamos el módulo de Contacts.

Sales From quotations to invoices Learn More	Restaurant Restaurant extensions for the Point of Sale Activate Learn More	Invoicing Invoices & Payments Learn More	CRM Track leads and close opportunities Learn More
ERP II Work Orders, Planning, Routing Learn More Upgrade	Website Enterprise website builder Activate Learn More	Inventory Manage your stock and logistics activities Learn More	Accounting Accounting, Taxes, Budgets, Assets... Learn More Upgrade
Knowledge Centralize, manage, share and grow your knowledge library Learn More Upgrade	Purchase Purchase orders, tenders and agreements Learn More	Point of Sale User-friendly PoS interface for shops and restaurants Activate Learn More	Project Organize and plan your projects Activate Learn More
eCommerce Sell your products online Activate Learn More	Manufacturing Manufacturing Orders & BOs Activate Learn More	Email Marketing Design, send and track emails Activate Learn More	Timesheets Track time & costs Learn More Upgrade
Expenses Submit, validate and reinvoice employee expenses Activate Learn More	Studio Create and Customize Applications Learn More Upgrade	Time Off Allocate PTOs and follow leaves requests Activate Learn More	Recruitment Track your recruitment pipeline Activate Learn More
Field Service Schedule and track onsite operations, time and material Learn More Upgrade	Employees Centralize employee information Activate Learn More	Data Recycle Find old records and archive/delete them Activate Module Info	Maintenance Track equipment and manage maintenance requests Activate Learn More
Sign Send documents to sign online Learn More Upgrade	Helpdesk Track support tickets Learn More Upgrade	Subscriptions MRR, Churn, Recurring payments Learn More Upgrade	Quality Quality Alerts, Control Points Learn More Upgrade

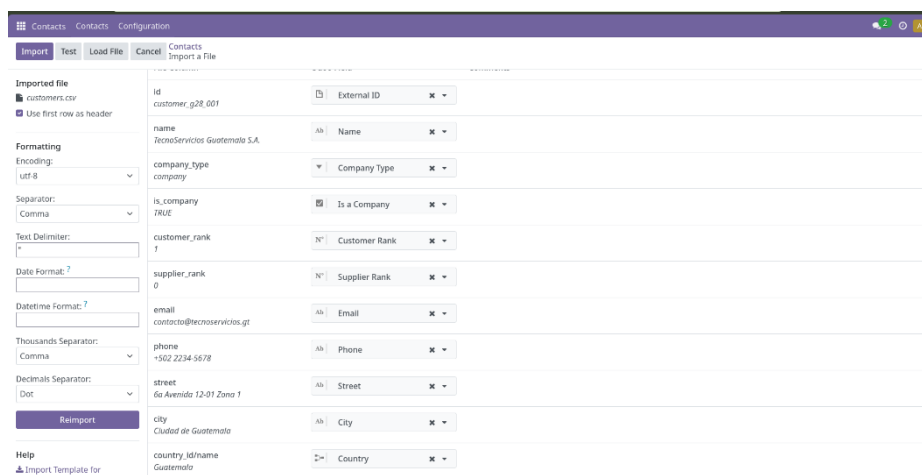
Y repetimos el mismo proceso que con los productos.



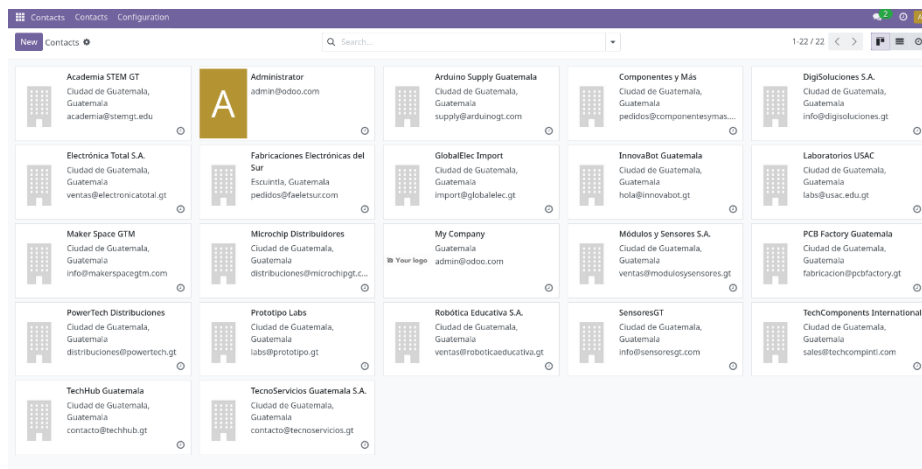
Debemos subir los archivos CSV de clientes y proveedores.



Verificamos que los clientes y proveedores se carguen correctamente.



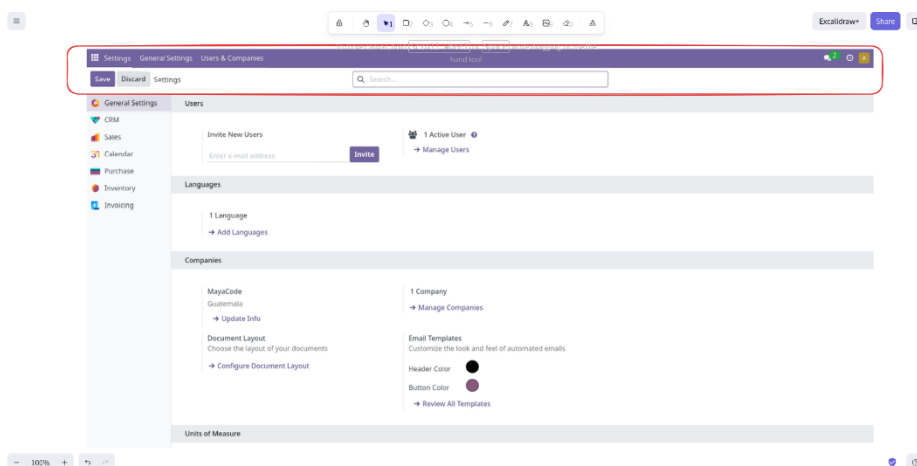
Verificamos que ambos archivos se hayan cargado correctamente.



CRM y ERP

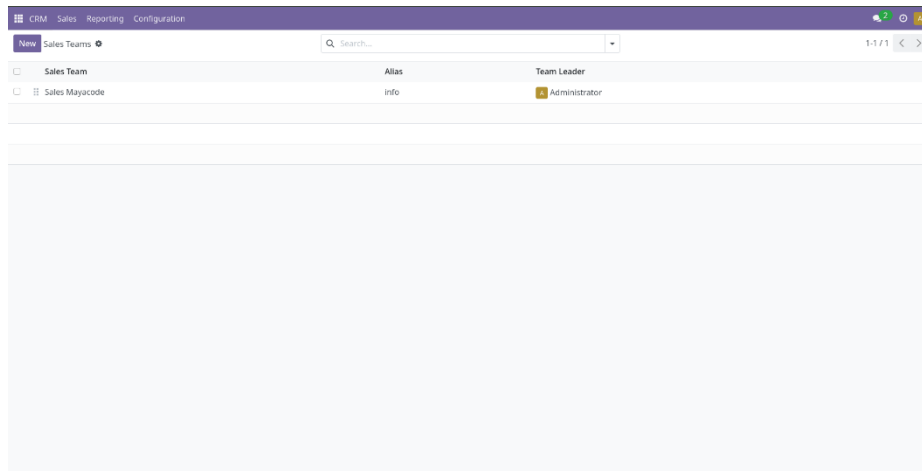
ERP (ventas, compras, inventario)

- Settings → Users & Companies → Companies — configura tu empresa (nombre, logo, dirección)
- Settings → Accounting — activa moneda (GTQ o USD)
- Inventory → Configuration → Warehouses — verifica que exista al menos 1 almacén



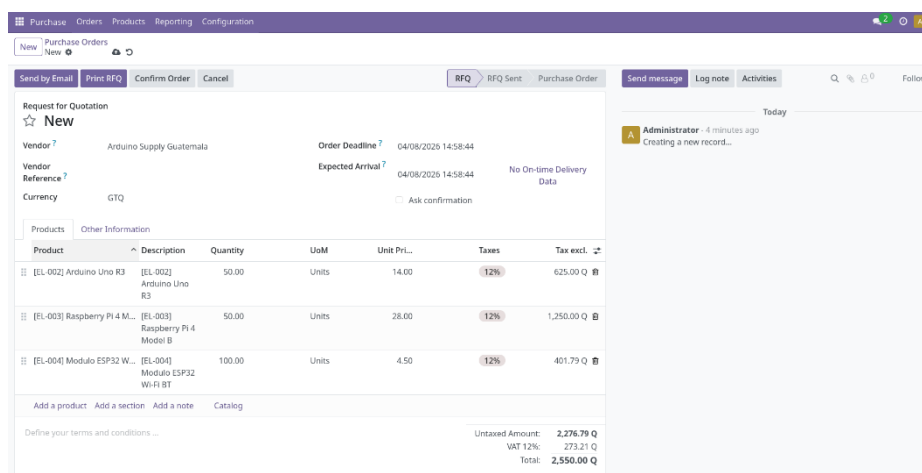
CRM

- CRM — configura etapas del pipeline (ej: Nuevo → Contactado → Propuesta → Ganado)
- CRM → Configuration → Sales Teams — crea un equipo de ventas



Compras a proveedores

- Ve a Purchase → Orders → Purchase Orders → New
- Selecciona proveedor (de los importados)
- En "Products": agrega productos, cantidad y precio
- Clic Confirm Order → genera una orden de compra oficial
- Cuando llegue la mercancía: Receive Products → valida el ingreso al almacén
- Print → genera la factura del proveedor



Purchase Orders / P00001

WH/IN/00001

1 / 1

ValidatePrintPrint LabelsCancel

DraftReadyDone

Send messageLog noteActivities

Follow

WH/IN/00001

Receive From: Arduino Supply Guatemala

Scheduled Date: 04/08/2026 14:58:44

Deadline: 04/08/2026 14:58:44

Source Document: P00001

OperationsAdditional InfoNote

Product	Demand	Quantity	Unit
[EL-002] Arduino Uno R3	50.00	50.00	Units
[EL-003] Raspberry Pi 4 Model B	50.00	50.00	Units
[EL-004] Modulo ESP32 Wi-Fi BT	100.00	100.00	Units

Add a line

Today

OdooBot - 47 seconds ago

This transfer has been created from: P00001

OdooBot - 47 seconds ago

Transfer created

Your logo

MayaCode
Guatemala

Vendor Address:
Arduino Supply Guatemala
Boulevard Vista Hermosa Zona 15
Ciudad de Guatemala
Guatemala
+502 5566-7788

Warehouse Address:
MayaCode
Guatemala
12345678

WH/IN/00001

Order: P00001

Shipping Date: 04/08/2026 15:05:45

Product	Ordered	Delivered
[EL-002] Arduino Uno R3	50.00 Units	50.00 Units
[EL-003] Raspberry Pi 4 Model B	50.00 Units	50.00 Units
[EL-004] Modulo ESP32 Wi-Fi BT	100.00 Units	100.00 Units

Cotizaciones de Productos

- Ve a Sales → Orders → Quotations → New
- Selecciona cliente
- Agrega productos en "Products"
- Clic Send by Email → envía al cliente / Confirm → convierte en orden de venta
- Clic en Create Invoice → genera la factura del cliente
- El flujo completo: Cotización → Orden de Venta → Entrega → Factura

The screenshot shows the 'New' quotation form in Odoo. The form is empty, showing fields for Customer, Expiration (05/08/2026), and Payment Terms. The Order Lines table is empty. The total is 0.00 Q.

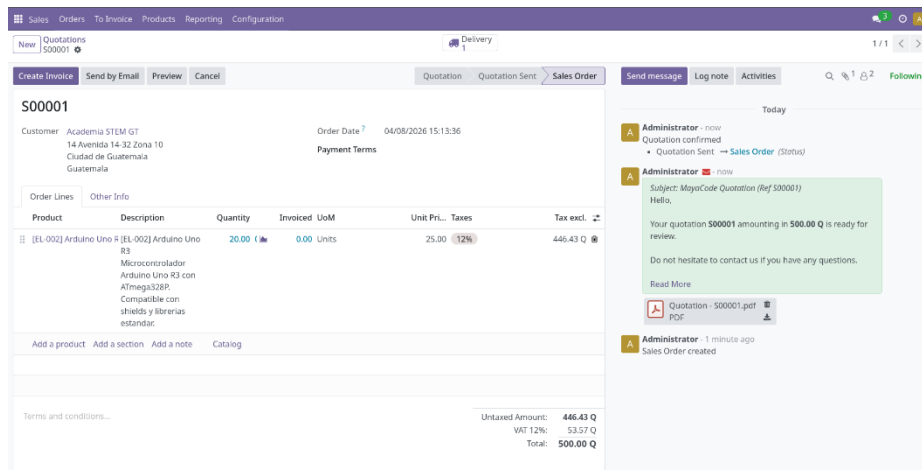
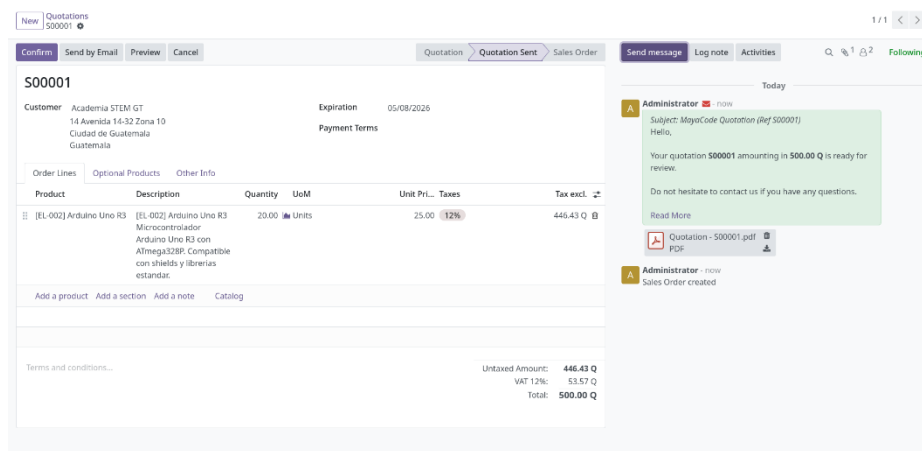
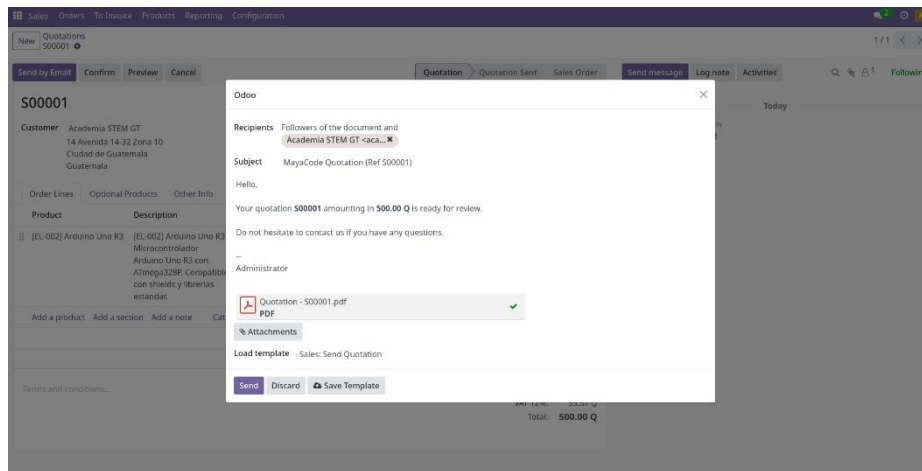
Product	Description	Quantity	UoM	Unit Pri...	Taxes	Tax excl.
Add a product Add a section Add a note Catalog						

Terms and conditions... Total: 0.00 Q

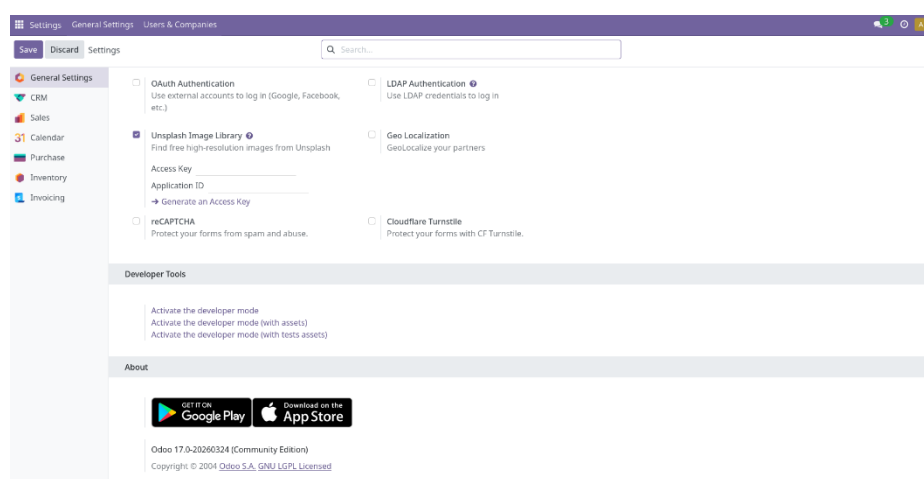
The screenshot shows the 'New' quotation form in Odoo with a product added. The product is 'Arduino Uno R3' with a quantity of 20.00. The total is 446.43 Q.

Product	Description	Quantity	UoM	Unit Pri...	Taxes	Tax excl.
[E]-002	Arduino Uno R3 Microcontrolador Arduino Uno R3 con ATmega328P. Compatible con shields y librerías estándar.	20.00	Units	25.00	(12%)	446.43 Q
Add a product Add a section Add a note Catalog						

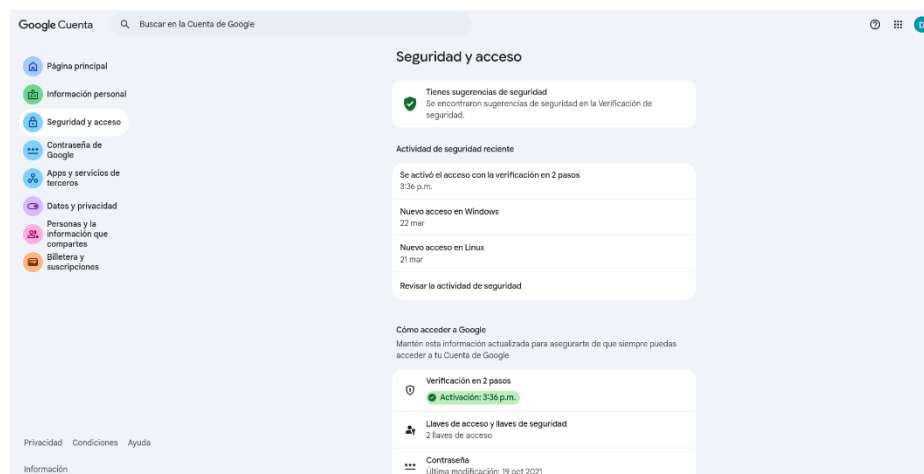
Terms and conditions... Untaxed Amount: 446.43 Q
VAT 12%: 53.57 Q
Total: 500.00 Q



Para enviar correos electrónicos, es necesario configurar un servidor de salida SMTP activando el modo desarrollador en **Settings → General Settings → Activate the developer mode**



Necesitamos Activar la verificación en 2 pasos del correo que vayamos a utilizar y generar una contraseña de aplicación.



Para crear la contraseña de aplicación accedemos al siguiente link:

<https://myaccount.google.com/apppasswords>

Google Cuenta

← Contraseñas de aplicaciones

Las contraseñas de la aplicación te permiten acceder a tu Cuenta de Google en apps y servicios más antiguos que no son compatibles con los estándares de seguridad modernos.

Las contraseñas de la aplicación son menos seguras que usar apps y servicios actualizados que cuentan con estándares de seguridad modernos. Antes de crear una contraseña de la aplicación, debes verificar si la app la necesita para acceder.

[Más información](#)

No tienes contraseñas de la aplicación.

Para crear una nueva contraseña específica de la app, escribe un nombre a continuación...

Nombre de la app
Ochoo

Crear

Privacidad Condiciones Ayuda Información

Configurar servidor de salida SMTP

Settings → Technical → Email → Outgoing Mail Servers → New
Ejemplo con Gmail

Si todo funciona bien debemos ver un mensaje de éxito a la izquierda:

New Outgoing Mail Servers

Gmail G28

Connection Test Successful

Test Connection

Name? Gmail G28 Priority? 10

FROM Filtering? e.g. email@domain.com, domain.com

Authenticate with? ☒ Username ☐ SSL Certificate ☐ Command Line Interface ☐ Gmail OAuth Authentication

Connect to your server through your usual username and password.
This is the most basic SMTP authentication process and may not be accepted by all providers.

Connection

Connection? ☐ None ☒ TLS (STARTTLS) ☐ SSL/TLS

Username? donheracademico@gmail.com

Password?

Debugging? ☐

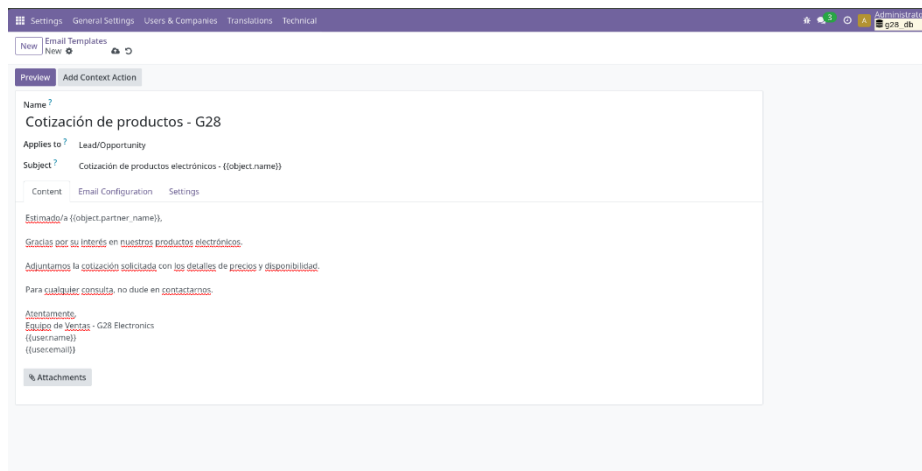
SMTP Server? smtp.gmail.com

SMTP Port? 587

Test Connection

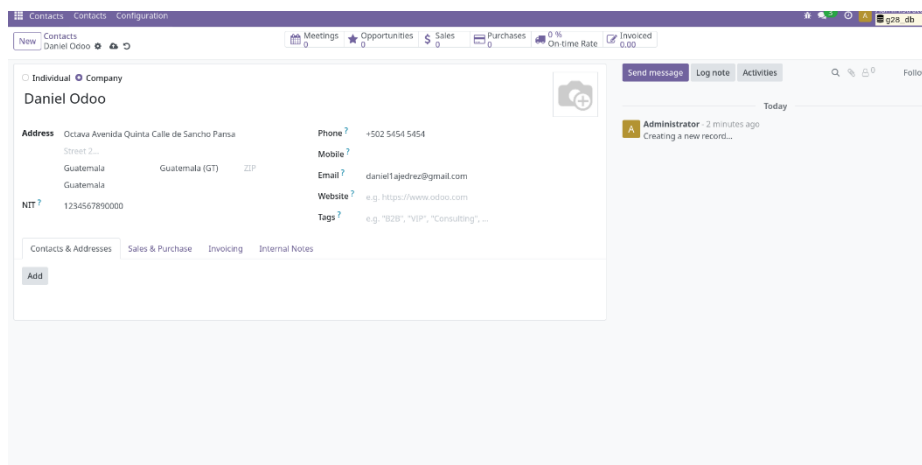
Vamos a crear una plantilla de correo electrónico para las cotizaciones:

- Crear una plantilla de correo
- Ve a Settings → Technical → Email Templates
- Clic New y llena:



The screenshot shows the 'Email Templates' configuration page in Odoo. The 'Name' field is 'Cotización de productos - G28'. The 'Applies to' field is 'Lead/Opportunity'. The 'Subject' field is 'Cotización de productos electrónicos - {{object.name}}'. The 'Content' tab is selected, showing a template with placeholders like {{object.partner.name}}, {{user.name}}, and {{user.email}}. There are tabs for 'Content', 'Email Configuration', and 'Settings'. An 'Attachments' button is at the bottom.

Tienes que tener a un usuario con correo electrónico configurado.



The screenshot shows the 'Contact' record for 'Daniel Odo'. The 'Address' field is 'Octava Avenida Quinta Calle de Sancho Pansa, Street 2...'. The 'Phone' field is '+502 5454 5454'. The 'Email' field is 'daniel1ajedrez@gmail.com'. The 'Website' field is 'https://www.odoo.com'. The 'Tags' field is 'e.g. "G28", "G28", "Consulting", ...'. There are tabs for 'Contacts & Addresses', 'Sales & Purchase', 'Invoicing', and 'Internal Notes'. An 'Add' button is at the bottom.

- Ve a CRM
- Abre o crea una oportunidad (clic New → ponle nombre y asigna un cliente)
- En el chatter (parte derecha), clic en Send message
- Aparece el compositor de correo → clic en el ícono "<->" y luego "Load a Template"
- Selecciona la plantilla que creaste → se autocompleta el asunto y cuerpo
- Verifica el destinatario (debe tener email el contacto)
- Clic Send

